TABLE 202-D AGGREGATE GRADATION VARIATION LIMITS

SIEVE SIZE	VARIATION LIMITS (% PASSING)
4.75 mm and larger	± 5
1.18 mm to 2.36 mm	± 3.5
0.300 mm and 0.600 mm	± 2
0.075 mm	± 1

202.20.03 Field Adjustment of Design Aggregate Gradation

The Contractor may make a maximum of two (2) field adjustments to the Design Aggregate Gradation, to accommodate minor changes in pit characteristics or to enhance production efficiency.

A field adjustment to the Design Aggregate Gradation is defined as a change in declared target gradation of the various aggregate sizes within the specified limits shown in Table 202-C which does not require a supplementary formal review. The maximum adjustment for individual sieve sizes for each adjustment are as permitted in the Table 202-E.

TABLE 202-E – FIELD ADJUSTMENT OF DESIGN AGGREGATE GRADATION

Sieve Designation (mm)	Maximum Field Adjustment (% Passing)
19 and larger	± 3.0
9.5	± 2.5
2.36 and 4.75	± 2.0
0.600 and 1.18	± 1.5
0.150 and 0.300	± 1.5
0.075	± 1.0

A proposed Design Aggregate Gradation field adjustment shall be submitted in writing with supporting documentation (showing original DAG, current stockpile average gradation, and proposed field adjustment and the new DAG) to the Quality Manager. Upon receipt of the proposed field adjustments, the Quality Manger will assess the adjustment for conformance with the contract requirements and notify the contractor whether or not it is acceptable in a timely fashion.

No field adjustment shall be acceptable if it results in a change from the current Design Aggregate Gradation which fall outside the limits defined in Tables 202-C. The variation limits specified in DB SS 202.20.02 shall apply to the adjusted DAG.

The Contractor shall have the moisture/density relationship established per ASTM test method D 698 for each original DAG, and one after each field adjustment thereto.

202.20.04 Declaration of a New Design Aggregate Gradation

Should there be a substantial change in the character of aggregate exposed in the pit face as the work proceeds, the Quality Manager may authorize a change in the declared Design Aggregate Gradation, which would then also be eligible for field adjustment in accordance with SS 202.20.

A maximum of one change to the declared DAG is to be permitted.

202.21 Stockpiling

202.21.01 All crushed aggregates shall be stockpiled prior to use.

202.21.02 Stockpile sites shall be cleared of all vegetation, trees, brush, rocks or other debris and a uniform gravel surface prepared before the stockpile material is deposited on the stockpile site.

202.21.03 Stockpiles shall be constructed on the designated site and when completed shall be neat and regular in shape, occupying as small an area as is practicable. Spilling of material over the edges of the piles shall not be permitted.

202.21.04 The Contractor shall ensure that stockpiles shall be built up in layers not to exceed 1 m in thickness.

202.21.05 The Contractor shall prevent dirt being tracked onto the crushed aggregate.

202.21.06 The Contractor may build the final stockpile by bulldozing the aggregate from a feed pile at the end of a production belt provided that the following are maintained:

i) The bulldozer(s) to be used shall be equipped with U-shaped pushing blades.

ii) The aggregate does not become contaminated with oversize material, mud or other objectionable material picked up from the pit floor or general working areas.

iii) The crushed granular aggregate is distributed evenly over the final stockpile area in lifts not greater than 150 mm in depth.

iv) The cone of the feeder pile at the end of the production belt shall not, without express permission, be allowed to build up to a greater height than 2 m.

v) No appreciable segregation or degradation shall occur in the main stockpile as a result of using this method.

PART F - CONSTRUCTION METHOD

202.22 Weather and Job Conditions

202.22.01 No construction shall be undertaken during snow, heavy rain, freezing or other unsuitable conditions. Aggregate shall not be placed upon a frozen, wet, muddy or rutted sub-grade, sub-base, base or surface.

202.22.02 When the sub-grade is soft due to excessive moisture, the placing of granular sub-base shall be stopped until rutting or displacement of the lower layers can be prevented by reduced loading or by other means.

202.23 Bridge End Fill - Construction of the bridge end fill shall not commence until the Designer has assessed the pertinent foundation conditions that may affect the future stability of both the bridge and embankments and has authorized continuation of construction.

The bridge end fill shall be laid out by extending back from the bridge abutment station, a distance of 8.0 m along centreline (and transversely parallel to the abutment) at finished grade and thence to a 1.5:1 backslope. The front slope, shall start at the bridge abutment station.

The bridge end fill shall be constructed to the subgrade elevation.

The material shall consist of mineral soil with properties and gradation in accordance with DBSS 202.04 and 202.05.

The bridge end fill shall be constructed in successive horizontal layers not exceeding 150 mm in loose thickness.

Each layer shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698.

The determination of field density shall follow a method approved by the Quality Manager.

202.24 Thickness of Granular Courses - Crushed surfacing course, crushed base courses and sub-base courses shall be constructed to the specified thickness and dimensions as shown on the design or as described in the Special Provisions.

Normally, on new construction, crushed bases shall be constructed 300 mm thick (in individual lifts with compacted thickness of 150 mm) unless otherwise approved by the Designer. Additional base or sub-base granular materials will be of such thickness as may be considered necessary to provide supporting strength for the flexible pavement structure.

202.25 Construction of 75 mm Sub-Base

202.25.01 Aggregates for sub-base shall be delivered to the roadbed as uniform mixtures and shall be spread in layers without segregation, preferably through an approved aggregate spreader. Granular aggregate shall not be end dumped from trucks in piles on the grade. The Quality Manager may permit spreading from the tailgate of trucks or from centre dump units, provided the Quality Manager is satisfied that the work will be well controlled and segregation will not occur.

When the sub-grade below granular aggregate sub-bases consists of cohesionless sand, and permission is granted by the Designer, a portion of the granular aggregate sub-base or base may be dumped in piles upon the sub-grade and spread ahead in sufficient quantity to stabilize the sub-grade. Segregation of aggregates shall be avoided and the material as spread shall be free from pockets of large or fine material. Segregated materials shall be remixed until uniform.

Where the required thickness is 150 mm or less, the granular base or sub-base may be spread and compacted in one layer. Where the required thickness is more than 150 mm, the aggregate shall be spread and compacted in two or more layers of approximately equal thickness; the maximum compacted thickness of any one layer shall not exceed 150 mm. Each layer shall be spread and compacted in a similar manner. The Contractor may lay a thicker lift, after demonstrating that the material can be thoroughly and uniformly compacted to the required density without degradation. At locations where the granular base or sub-base is to be placed over areas inaccessible to the spreading equipment, the granular sub-base or base may be spread by any means to obtain the specified results.

202.25.02 Immediately following spreading, the material shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698.

If, because of the nature of the granular aggregate, the Quality Manager considers that the results from the above test will be unreliable or the use of the test for density measurements is impractical, then the Quality Manager may order that each lift or course of aggregate shall be continuously and thoroughly rolled until successive passes of a vibratory roller results in an increase in density of less than 10 kg/m^3

So that development of new compaction equipment will not be inhibited, the Quality Manager may give permission for the substitution of new or alternative compaction equipment if the Quality Manager is satisfied that such equipment will provide equal or superior compaction performance.

202.25.03 Subject to the concurrence of the Quality Manager, the Contractor may water the sub-base aggregates as required to aid in attaining the specified density.

202.25.04 The completed surface of the Sub-base course shall conform to the required line, grade and cross section as shown on the Design to an accuracy ± 15 mm, neither uniformly high nor low.

202.26 Construction of 25 mm Crushed Base Course

202.26.01 Crushed 25 mm base course aggregate shall be placed on a properly prepared Sub-Base surface to such depth or at such rates as may be specified. If the Quality Manager is of the opinion that the finished surface of the Sub-Base does not meet the requirements of DBSS 202.25.04 but has been thoroughly and densely compacted and should not be disturbed, the Quality Manager may order that the surface of the Sub-Base be corrected to the true cross section, line and grade, and within the tolerances specified by use of a levelling course of crushed 25 mm base course aggregate.

The 25 mm crushed base course shall be constructed in such a manner that the aggregate is neither segregated, contaminated nor degraded. End dumping will not be permitted. The thickness of the 25 mm crushed base course shall be substantially uniform and the minimum thickness shall not be less than the nominal thickness shown on the Design or ordered by the Designer. If the Contractor is unable to provide adequate manually operated equipment or workers of sufficient skill to lay the 25 mm crushed base course aggregate within the tolerances specified, the Contractor shall lay the aggregate through an approved electronically controlled spreading machine.

202.26.02 Compaction - Immediately following spreading, the 25 mm crushed base course aggregate shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698.

The method of compaction to be employed may be selected by the Contractor, but shall be subject to approval by the Quality Manager.

202.26.03 Watering - If required, water shall be applied to attain the specified density.

202.26.04 The completed surface of the granular base course shall conform to the required line, grade and cross section as shown on the Design to an accuracy ± 10 mm, neither uniformly high nor low.

202.27 Equipment for Watering - Water shall be applied from a distributor of the pressure type, equipped with a spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water accurately and uniformly. Splash plate type distributors or those equipped with spray bars that eject fine streams of water will not be permitted. The distributor must be provided with a satisfactory means for accurately measuring the quantity of water sprayed. If the Quality Manager so requires, measuring equipment shall be calibrated under the Quality Manager's inspection.

202.29 Proof Rolling and Stabilization- Before acceptance, each compacted course of base course aggregate shall receive one complete coverage by the tires of a truck having a 9 tonne single axle dual tire or 17 tonne tandem axle group with dual tires with a tire pressure of 600 kPa.

Any areas where rutting or displacement occurs shall be either excavated and replaced or stabilized by the addition of suitable blending material incorporated uniformly into the base.

Blending may be performed at the pit or quarry or on the highway in whichever manner the Quality Manager may direct and to the satisfaction of the Ministry Representative.

202.30 Benkelman Beam Testing - At any time during the course of the work, when considered necessary by the Quality Manager or the Ministry Representative, Benkelman Beam testing may be undertaken.

202.31 Water Applied to the Highway - Water for compaction or dust control shall be incidental to the Contract.

202.32 Development and Cleanup of Ministry-Provided Pits and Quarries

202.32.01 Authorized clearing and grubbing of Ministry-provided pits and quarries, and construction of the access road is the responsibility of the Contractor. Removal and stockpiling of overburden and topsoil in the development of the proposed work areas, and spreading overburden to reshape the sides of a pit and spreading topsoil over a depleted pit or quarry shall be the responsibility of the Contractor and the work shall be completed to the satisfaction of the Ministry Representative.

202.32.02 Pit and quarry development work, restoration work, and construction of access road will be paid for by Order for Extra Work where, in the opinion of the Ministry Representative:

i) Not Used

ii) The gravel pits or quarries are designated subsequent to tendering.

iii) Development and restoration work is substantially different from that described in the Contract.

iv) Restoration work is required outside the current work area.

v) Not Used

202.33 Stockpile Reject Aggregate

202.33.01 Aggregate from screening or crushing operations in a Ministry-provided pit or quarry, whether:

i) Oversize under DBSS 202.09, or

ii) Rejected at the crusher in order that the product meets the requirements of DBSS 202.05,

shall be stockpiled separately according to size in accordance with DBSS 202. Stockpiles of reject aggregate shall not be contaminated by organic or other deleterious materials.

202.33.02 No payment will be made for any reject aggregate.

202.34 Blending Materials for Stabilizing Aggregates on the Highway - Blending materials, supplied, loaded, hauled, placed and mixed into the granular aggregate on the highway to correct deficiencies in aggregate stability, shall be the responsibility of the Contractor.

202.35 Not Used

202.36 Aggregates Supplied in Place

202.36.01 Not Used

202.36.02 Not Used

202.36.03 Surplus Aggregate

Surplus aggregates in stockpile on Ministry lands are the property of the Ministry, and no additional compensation will be paid.

Appendix 1

Fracture Count for Coarse Aggregate
(BCH 1-13)

Purpose:

This test determines the amount of fractured material in the coarse part of an aggregate sample.

Method "A" is used for crushed granular surfacing and base aggregates.
Method "B" is used for paving aggregates.

Apparatus:

1. Drying oven, preferably forced draft, capable of maintaining a uniform temperature of 110 $\pm$ 5 ^{o} C.
2. A nest of sieves of the following sizes:
   Method "A": 37.5 mm, 25.0 mm, 19.0 mm, 12.5 mm, 9.5 mm, 4.75 mm
   Method "B": 19.0 mm, 13.2 mm, 9.5 mm
3. Balance with sufficient capacity and accurate within 0.1 percent of the mass of sample.

Test Sample:

The sample of aggregate to be tested shall be representative, oven-dry, and reduced to an amount suitable for testing. The mass of the sample shall conform to the following:

Maximum Nominal Size (mm)	Minimum Dry Mass (kg)
37.5	4.0
25.0	2.5
12.5	1.5
9.5	1.0

Procedure: Method "A"

1. Sieve the coarse aggregate to be tested on the 37.5 mm, 25.0 mm, 19.0 mm, 12.5 mm, 9.5 mm 4.75 mm sieves.
2. Separate each sieve size into two groups, fractured and unfractured. Each rock in the fractured group shall have at **least one** fractured face or shall be naturally angular with sharp edges. (See Discussion 1).
3. For each sieve size, count the number of fractured rocks and the total number of rocks (fractured and unfractured).
4. Calculate and record the % fracture for each sieve size and report the total % fracture for the sample using the following formulas:

FOR EACH SIEVE SIZE: (i.e. 19 mm)

$$\% \text{ Fracture (19 mm)} = \frac{\text{Number of Fractured Rocks (19 mm)}}{\text{Total No. of Rocks (19 mm)}} \times 100\%$$

FOR THE TOTAL SAMPLE:

$$\text{Total \% Fracture} = \frac{\text{Number of Fractured Rocks (All Sieve Sizes)}}{\text{Total No. of Rocks (All Sizes)}} \times 100\%$$

Procedure: Method "B"

1. Separate the coarse aggregate to be tested from the fine aggregate by sieving it over the 4.75 mm screen.
2. Separate the coarse aggregate for each sieve size into two groups, fractured and unfractured. Each rock in the fractured group shall have **two or more** fractured faces produced by a crusher or shall be naturally angular with sharp edges. Each rock in the unfractured group shall be unfractured or fractured only in one dimension.
3. Obtain the mass of the fractured group, also the mass of the total coarse aggregate (fractured + unfractured groups).
4. Calculate and report the total % fracture for the sample using the following formula:

$$\% \text{ Fracture} = \frac{\text{Mass of Fractured Rock}}{\text{Mass of Total Rocks}} \times 100\ \%$$

Discussion:

-Fractured face shall be defined as a fracture in any plane whose areas is 15% or more of the largest cross-section area in a parallel plane.
-The required sample can be conveniently obtained from the separated fractions of the Dry or Wash Sieve Test.

SECTION 204

ROCK CUTS

DESCRIPTION

204.01 Scope - This Section describes the controlled blasting techniques required to produce rock cut slopes at the locations shown on the Design or as required by the Contract. Rock shall be excavated to subgrade elevation as shown on the Design. The work shall be conducted, whether by blasting or other methods, in a manner that minimizes blast damage to the excavation backslope and minimizes the requirement for stabilization. The responsibility for the results of every blast lies with the Contractor. Nothing within this specification shall be interpreted to mean the Ministry accepts responsibility for the results of any blast.

204.02 Definitions

204.02.01 Backline Holes - A line of holes detonated or intermittently detonated along the specified excavation backslope.

204.02.02 Bench – A near-horizontal or shallow slope surface at the top of a near-vertical rock face where blast holes are collared.

204.02.03 Bench Width – The horizontal distance on the bench between the excavation backslope and the crest of the existing rock face. Generally this dimension defines the width of the rock to be removed and is perpendicular to the highway alignment.

204.02.04 Blast Crater - A local depression in rock formed by blasting.

204.02.05 Buffer Holes - The line of production holes located closest to and parallel to the backline holes.

204.02.06 Burden - The distance between the blasthole and the nearest free face.

204.02.07 Charge - A specific quantity of explosive such as a cartridge.

204.02.08 Controlled Blasting - Controlled blasting is the controlled use of explosives and accessories in carefully spaced and aligned production and backline drill holes to produce the specified excavation backslope within the specified excavation limits. The purpose of controlled blasting is to minimize damage to the rock backslope, to help ensure long-term stability and to minimize flyrock.

204.02.09 Controlled Delay Sequence - The delay sequence required for controlled blasting.

204.02.10 Cushion Blasting - A blasting method whereby the production holes are detonated before the backline holes.

204.02.11 Delay - A blast initiator with a specific delay time period before detonation.

204.02.12 Excavation Backslope or Smooth Wall - This is the location of the free surface or shear plane in the rock produced by blasting along the backline holes.

204.02.13 Excavation Limits - This is the boundary within which rock is removed.

204.02.14 Explosive Decoupling - The separation of an explosive charge from the wall of the blast hole.

204.02.15 Flyrock - The throw of fragmented material during blasting.

204.02.16 Free Face - The rock face that provides relief for a blast.

204.02.17 Guide Holes - Those holes along the backline that are not loaded with explosive and are not stemmed.

204.02.18 Hole Deviation – Borehole misalignment distance measured in two directions, within and perpendicular to the excavation backslope. Within the plane of the excavation backslope, hole deviation is borehole misalignment as measured by the horizontal offset distance between any point in the design hole location and the point at a corresponding depth in the actual hole. Perpendicular to the plane of the excavation backslope, hole deviation is the horizontal offset distance between the plane of the design excavation backslope and the centreline of the borehole being measured.

204.02.19 Lift - The vertical distance between the top and bottom of an area to be blasted.

204.02.20 Overbreak in Rock - Overbreak in rock is any rock outside of the specified excavation limits (except slide material described in DBSS 201.42) that is excavated, displaced or loose due to the inherent character of any formation encountered or due to any other cause.

204.02.21 Overburden - This is Type B, C and/or D material (as described in DBSS 201.11) overlying rock.

204.02.22 Presplit (or Pre-Shear) Blasting - A blasting method whereby backline holes are detonated before production holes are detonated.

204.02.23 Production Blasting - Blasting of production holes detonated in a controlled delay sequence.

204.02.24 Production Holes - All holes within the specified excavation limits that are not backline holes. Production holes are often more widely spaced than backline holes.

204.02.25 Rock Excavation - Rock excavation includes drilling holes, loading holes with explosives, detonation by controlled blasting, mucking to subgrade and ditch level, and removal of material to a designated area. Rock excavation also includes secondary breakage of oversize rock.

204.02.26 Rock - Rock is Type A material as defined in DBSS 201.11.01.

204.02.27 Scaling - The removal of all loose material using scaling bars, portable hydraulic jacks, other hand tools, wire rope cables, compressed air blow pipes, blasting without the need for drilled holes and other methods authorized by the Ministry Representative.

204.02.28 Setback Distance - The setback distance is the horizontal distance between the top of the excavation backslope and the toe of the overburden material. Setback is created by removing overburden.

204.02.29 Spacing - The distance between blastholes perpendicular to the burden.

204.02.30 Stemming - Material placed on top of the explosive to provide confinement of the explosive gases.

204.02.31 Subdrill - The distance drilled below subgrade level.

204.02.32 Trimming - The removal by drilling and controlled blasting of potentially unstable rock remaining at the exposed excavation backslope. Trimming may not require backline holes.

204.03 Quality Management

204.03.01 General - Quality management shall be conducted for all blasting operations.

204.03.02 Blasting Consultant - The Contractor shall retain a specialist blasting consultant, acceptable to the Quality Manager , to provide quality control. The consultant shall not be an employee of the Contractor, explosives manufacturer or explosives distributor.

204.03.03 Vibration Specialist - The Contractor shall retain a vibration specialist to provide blast vibration monitoring if required. The specialist shall not be an employee of the Contractor, explosives manufacturer or explosives distributor.

204.03.04 Quality Management Activities For Blasting Operations - Quality management for blasting operations shall be performed for test sections, full scale blasting operations and trimming. A field report, generated by the Consultant shall be provided to the Quality Manager, and upon request to the Ministry Representative, within one day of each site visit by the Consultant.

Quality control for blasting includes, but is not limited to the following activities by the blasting consultant:

- Viewing the proposed blast area at least one (1) week before drilling operations commence.
- Preparing and submitting blast designs for production and backline holes.
- Intermittently viewing the site during drilling for each blast area.
- Intermittently observing the loading of holes with explosives and tying in to the delay system.
- Observing the blasts and reviewing the excavated areas.
- Attending on site to view the excavation backslope after each lift and reviewing excavation backslope for quality control purposes.
- Viewing the site while reviewing proposed changes to the blast design and preparing new designs.
- Attending on site at other times as are appropriate to assist the Contractor and Quality Manager in the setting up, assessment and adjustment of the various procedures to be employed for blasting.
- Preparing a field report for each site visit, including details of the progress of blasting operations, a statement whether the blast design is being complied with, any changes to the blast design, any recommendations made to the Contractor and Quality Manager and any problems encountered by the Contractor and Quality Manager.

204.04 Submittals

204.04.01 General – The Ministry will perform Quality Audit. The receipt of submittals by the Ministry shall not relieve responsibility from the Contractor for the accuracy and adequacy of the submittals. Submittals are for quality audit and record keeping purposes. Inadequate or incorrect submittals will be returned for revisions prior to acceptance. The Contractor shall submit to the Quality Manager , and upon request to the Ministry Representative, the following documentation for review.

204.04.02 Qualifications and Experience of Contractor - The Contractor shall provide a statement of the qualifications, experience and work function of all personnel assigned to drilling and blasting duties. A statement of previous work experience on similar projects shall also be provided. This statement shall include the project name, location, volume of rock, year constructed and the owner/client contact name. The company, the driller and the blaster shall each have a minimum of 5 consecutive years demonstrated experience in drilling and controlled blasting work on at least 3 projects involving rock cuts over 8 m height along transportation corridors.

204.04.03 Qualifications and Experience of Proposed Blasting Consultant - The Contractor shall provide the name of the consulting firm, and the name and qualifications of the blasting consultant's on-site representative who will be providing the quality control for rock excavation. The blasting consultant shall have a minimum of 5 consecutive years demonstrated experience in preparing successful blast designs along transportation corridors for at least 3 projects. The following information shall be included in the qualifications submitted:

- Project name, location and experience.
- Name and phone number of owner/client contact who can verify the experience of the blasting consultant's site representative.

204.04.04 Qualifications and Experience of Proposed Vibration Specialist - The Contractor shall provide the name and qualifications of the vibration specialist who will be providing the quality control for rock excavation. The vibration specialist shall have a minimum of 5 consecutive years demonstrated experience in the field of vibration monitoring for at least 3 projects. The following information shall be included in the qualifications submitted:

- Project name, location and experience.
- Name and phone number of owner/client contact who can verify the experience of the vibration specialist.

204.04.05 Proposed General Construction Plan - The Contractor shall submit a general construction plan to the Quality Manager showing proposed blasting sequence numbers defining the order of blasts for the contract. The plan shall also show the following information for each blast:

- Blast sequence number.
- The location of the blast in relation to the remaining rock on the site.
- Approximate dimension of the rock to be removed.
- Estimated volume of rock to be removed.
- Location of the disposal site.

204.04.06 Sequence of Operations – The Contractor shall adopt a logical, systematic sequence of operations to ensure blasting is conducted safely and effectively. The following general sequence of blasting operations outlines minimum requirements of the Contractor to maximize the stability of the excavation backslope and does not limit or supercede any other requirements in this specification:

- Retain a blasting consultant to perform quality control.
- Submit blast designs for quality audit purposes to the Ministry Representative.
- Accurately survey locations of proposed blast holes.
- Drill holes, load explosives, detonate blast and remove muck pile.
- Check the excavation backslope to determine hole offsets, hole angles, hole alignment and compliance with allowable tolerances.
- Review blast results and make changes to blasting operations as necessary.
- Perform backslope stabilization to the satisfaction of the Quality Manager before subsequent lifts are detonated.

204.04.07 Blast Design - The Contractor shall provide and follow a blast design, approved and signed by the blasting consultant, not less than one week prior to commencing drilling and blasting operations and a minimum of one day before the Contractor proposes to implement any changes to the previously utilized drilling or blasting methods. The design may be prepared by the blaster, but shall be reviewed by the Contractor and forwarded to the blasting consultant for signing. The design shall contain full details of the drilling and blasting patterns and controls that the Contractor proposes to use for controlled blasting. The blast design shall contain the following minimum information:

- Date the design was prepared and proposed date of blast.
- Station limits of proposed blast.
- Plan and section views of proposed drill pattern including free face, burden, blasthole spacing, blasthole diameters, blasthole angles, lift height, hole depth, and subdrill depth.
- The location of production (including buffer) and backline holes.
- Loading diagram showing type and amount of explosives, primers, initiators, and location and depth of stemming.
- Initiation sequence of blast holes including delay times and delay system.
- Manufacturer's data sheets for all explosives, primers, delays, and initiators to be used.
- Blasting consultant's signature, printed name, and company name.
- Blaster's signature, printed name, company name, and blaster's certificate number.

204.04.08 Preblast Survey - The Contractor shall conduct a preblast survey a minimum of one day before blasting operations commence. The preblast survey shall include a complete description of the existing condition of any nearby buildings, structures, wells and utilities that potentially may be damaged by blasting operations. The survey method used shall be acceptable to the Contractor's insurance company.

204.04.09 As-Built Blast Design - Within one day after each blast, the Contractor shall submit to the Quality Manager, an as-built blast design showing all actual blast details in a format that permits direct comparison with the proposed blast design.

204.04.10 Blasting Consultant's Field Report - The Contractor shall provide the blasting consultant's field report to the Quality Manager within one day after each visit by the blasting consultant.

204.04.11 Vibration Control Records - As required, the Contractor shall provide all seismograph records of vibration monitoring and interpretation of results within one day after each blast to the Quality Manager.

204.04.12 Peak Overpressure Records - As required, the Contractor shall provide a permanent signed and dated record of the peak overpressure measurements within one day after each blast to the Quality Manager.

MATERIALS

204.11 Explosives and Related Products

204.11.01 Manufacturer - All products and materials used for any blasting operations shall be products of a company regularly engaged in the manufacture of explosives and related products.

204.11.02 Expired Explosives - Explosives with an expired shelf life shall not be used.

204.11.03 Water Resistance - Water resistant explosives may be required for the work.

CONSTRUCTION

204.31 Permits and Regulations - The Contractor shall obtain all necessary permits for and shall comply fully with the laws, rules and regulations of Municipal, Provincial and Federal agencies in connection with the use, transportation, storage and safe handling of all explosives, including those regulations contained in the Industrial Health and Safety Regulations (IHSR) published by the Worker's Compensation Board (WCB) of British Columbia.

204.32 Supervision - The Contractor shall provide at least one person thoroughly trained and experienced in the use of explosives who shall be present at all times during the execution of all blasting operations and who shall direct such work.

204.33 Personnel - The Contractor shall ensure that all persons conducting blasting operations have a valid blaster's certificate issued by the WCB or is under the direct supervision of a certificate holder.

204.34 Safety - The Contractor shall meet all WCB regulations. All work shall be performed in a manner that prevents injury or harm to any personnel employed in the rock excavation area. Warning signs shall be posted and readily recognizable audible warning signals shall be used. The perimeter of the area affected by blasting operations shall be patrolled and controlled by direct voice communication. The Quality Manager shall stop work if the safety of the public is being jeopardized by the Contractor's blasting operations.

204.35 Flyrock Control - Before the detonation of any blast in areas where flying rock or other debris may result in personal injury or damage to property, the area within the excavation limits shall be covered with suitable blasting mats, soil or other equally serviceable material to prevent flyrock.

204.36 Other Damage - The Contractor shall be responsible for any damage resulting from blasting. Occupants of local buildings shall be notified by the Contractor, prior to the commencement of the blasting, as to the timing, size of blasts, types of warning and other signals.

204.37 Subgrade and Ditch Areas

204.37.01 General - The subgrade shall be constructed to ±50 mm of the specified line and grade before the placement of Select Granular Sub-Base Material. Any pinnacles of intact rock protruding above the design subgrade elevation shall be removed.

204.37.02 Supporting Rock - The Contractor shall exercise care and use appropriate methods to prevent breaking, loosening or otherwise damaging supporting rock below subgrade level and ditch bottom. The Contractor shall be responsible for the methods used and for any damage to the rock structure resulting from the operations.

204.37.03 Drainage - Excavated rock areas shall be free draining. If craters formed in rock by blasting below subgrade elevation are not free draining, then the Contractor shall provide drainage by trenching to a free outlet. These blast craters and drainage trenches shall be backfilled to subgrade elevation.

204.37.04 Shatter Zone for Drainage -Drainage shall also be provided by shattering the upper portion of rock below subgrade and ditches within the entire excavation limits. The shatter zone shall be 300 mm thick. A shatter zone thickness of 900 mm may be required near cut to fill transition zones for 10 m beyond the transition.

204.37.05 Backfill Material - Backfill shall be free draining material such as clean broken rock or coarse clean granular material. All backfill shall be placed and compacted as specified.

204.37.06 Ditches - The ditches shall be formed and cleaned before any base material is placed on the subgrade.

204.38 Blasting Test Section(s)

204.38.01 General - All requirements for full scale blasting shall also apply to test sections. Prior to commencing full-scale blasting operations, the Quality Manager may require the Contractor to demonstrate the adequacy of the proposed blast design by drilling, blasting, and excavating short test sections, up to 30 m in length, to determine which combination of method, hole spacing, timing, and charge yields acceptable results. The length of the blast test section shall be as specified in the blast design.

204.38.02 Reduced Section Length - Where requested by the Quality Manager, the Contractor shall use test section lengths of less than 30 m.

204.38.03 Backline Hole Spacing - The Contractor shall begin the tests by drilling backline holes 750 mm apart

along the specified excavation backslope, then adjust if needed, until the Quality Manager authorizes the spacing to be used for full-scale blasting operations.

204.38.04 Review of Test Section Results - The Contractor shall not drill beyond the test section until it has been excavated and the results reviewed by the blasting consultant and continuation is authorized by the Quality Manager.

204.38.05 Revision of Methods - If either the blasting consultant or the Quality Manager determines that the results of the test section are unsatisfactory, then the Contractor shall adopt such revised methods as are necessary to achieve the required results.

204.38.06 Additional Test Sections - If at any time during the progress of the work the methods of drilling and blasting do not produce the required excavation backslope geometry within the tolerances specified for backline holes, then the Contractor will be required to drill, blast and excavate short sections, not exceeding 30 m in length, until a technique is achieved that will produce the desired results.

204.39 Overburden Removal and Setback - The setback distance shall be 3.0 m. The Contractor shall, before drilling the backline holes, remove all overburden within the excavation limits or 10 m beyond the limits of the production holes in a direction parallel to the backline. The overburden surface shall be sloped at 1.5:1**204.40 Backline and Production Holes**

204.40.01 Stemming - The upper portion of all holes between the topmost charge and the hole collar shall be stemmed. Stemming materials shall be sand or other inert angular granular material with similar specific gravity passing a 9.5 mm sieve.

204.40.02 Hole Obstructions - Before placing charges, the Contractor shall determine that the hole is free of obstructions for its entire depth. All necessary precautions shall be exercised so that the placing of the charges will not cause spalling of material from the walls of the holes.

204.40.03 General Orientation - All holes shall be drilled downward unless otherwise specified in the blast design. In general, slash holes (horizontal, near horizontal or fanned out holes) shall not be drilled along the excavation backslope or on pioneering routes excavated to provide access for backline hole drilling.

204.40.04 Insertion of Explosive - Explosive materials shall not be inserted into the holes until the blast design has been reviewed by the Quality Manager.

204.41 Backline Holes

204.41.01 General - The Contractor shall control the hole layout and drilling operations using proper equipment and techniques to ensure that backline holes are located and oriented correctly. The proposed location of each backline hole shall be accurately surveyed and staked.

204.41.02 Hole Location - For each blast, the line of backline holes shall extend 10 to 20 m beyond the limits of the production holes to be detonated or to the end of the specified excavation backslope, as applicable.

204.41.03 Hole Location Tolerance - Backline holes shall be drilled within 75 mm of the staked collar location.

204.41.04 Hole Deviation - Backline holes shall not deviate from the plane of the specified excavation backslope by more than 150 mm as measured perpendicular to the slope. Backline holes shall not deviate more than 150 mm as measured within the plane of the excavation backslope

204.41.05 Hole Diameter - The backline holes shall be 50 to 75 mm in diameter, or as recommended by the blasting consultant. Trim blast holes may be as small as 25 mm diameter.

204.41.06 Hole Length - The length of backline holes for any individual lift shall not exceed 8 m unless the Contractor can demonstrate to the Quality Manager that the Contractor can stay within the tolerances and produce the required excavation backslope geometry.

204.41.07 Guide Holes - Guide Holes shall be of the same diameter and drilled in the same plane and to the same tolerance as the remaining backline holes.

204.41.08 Hole Length Increase - Upon satisfactory demonstration, the length of holes may be increased to a maximum of 12 m with written authorization of the Quality Manager. If more than 5% of the backline holes are misaligned in any one lift, then the Contractor shall reduce the height of the lifts until the 150 mm alignment tolerance is met.

204.41.09 Control of Drilling Orientation - All drilling equipment used to drill the backline holes shall have mechanical devices attached to that equipment to accurately determine the orientation of the drill steel entering the rock. Backline hole drilling will not be permitted if these devices are either missing or inoperative.

204.41.10 Offset Between Lifts - When the cut height requires more than one lift, a maximum 0.5 m offset between lifts is permitted to allow for drill equipment clearances. The Contractor shall begin the backline hole drilling at a point on the top lift which will allow for necessary offsets and shall adjust at the start of lower lifts to compensate for any drift which may have occurred in the upper lifts.

204.41.11 Length for Toe Berm Removal - Drilling 0.5 m below ditch bottom is permitted to facilitate removal of the toe berm.

204.42 Presplit Blasting

204.42.01 General – Unless otherwise specified in the blast design , presplit blasting shall be conducted for all blasting to the excavation backslope. In general, cushion blasting may proceed when the bench width is less than three times

the lift height.

204.42.02 Explosive Type and Accessories - Drill hole conditions may vary from dry to filled with water. The Contractor shall use explosives and blasting accessories appropriate for the drill hole conditions encountered to accomplish the specified results. Only standard explosives manufactured for presplit blasting shall be used in backline holes, unless otherwise specified in the blast design. Bulk ammonium nitrate and fuel oil (ANFO) shall not be loaded into the backline holes.

204.42.03 Explosive Decoupling - Explosives shall be evenly distributed and decoupled to the maximum extent possible.

204.42.04 Explosive Charges - The bottom charge of backline holes may be larger than the remaining charges but shall not be large enough to cause overbreak. The top charge of backline holes shall be placed far enough below the collar and be sufficiently small to avoid overbreak and heaving of rock beyond the excavation backslope.

204.42.05 Modified Blasting Sequence - The Contractor may detonate the backline holes before drilling production holes, provided satisfactory excavation backslopes are obtained.

204.42.06 Backline Hole Delays - If required to reduce ground vibrations or noise, backline holes may be delayed, provided the effective hole-to-hole delay time is not more than 25 ms.

204.42.07 Excavation Backslope Geometry - The excavation backslope shall not deviate more than 150 mm from a plane passing through adjacent drill holes except where the character of the rock is such that irregularities are unavoidable. The 150 mm tolerance shall be measured perpendicular to the plane of the slope. In no case shall any portion of the slope encroach on the ditch.

204.43 Cushion Blasting

204.43.01 General - Where the horizontal distance from the specified excavation backslope to the existing free face is less than 5 m, the Contractor may use cushion blasting instead of presplit blasting.

204.43.02 Delay Time - The difference in delay time between the backline holes detonated after the buffer holes shall be between 25 and 75 ms. With the exception of these criteria, requirements for presplit blasting shall also apply to cushion blasting.

204.44 Production Blasting

204.44.01 Hole Diameter - Production holes shall not exceed 150 mm in diameter, unless otherwise specified in the blast design.

204.44.02 Hole Location - Production blast holes (not including buffer holes) shall not be drilled closer than 2.0 m to the backline holes, unless otherwise specified in the blast design. The bottom of the production holes (including buffer holes) shall not be lower than the bottom of the backline holes.

204.44.03 Delay Sequence - All sequences shall be delayed in such a manner that successive delays promote the movement of rock in the direction of a free face at all times.

204.44.04 Damage to Backslope - It is the Contractor's responsibility to take all necessary precautions during production blasting to minimize blast damage to the excavation backslope.

204.44.05 Buffer Holes - A line of buffer holes shall be drilled along a plane parallel to the backline holes. Buffer hole diameters shall be between 50 and 75 mm unless otherwise specified in the blast design. Unless otherwise specified in the blast design, the line of buffer holes shall be drilled approximately 1 m away from the backline holes and spaced 1.0 to 1.5 m centre to centre.

204.44.06 Buffer Hole Charge and Detonation - The explosive charge in buffer holes shall not exceed 50% of the full explosive load that could be placed in a 75 mm diameter production hole. Detonation of the buffer holes shall be on a delay sequence toward a free face. Ammonium nitrate and fuel oil shall not be used for buffer holes unless otherwise specified in the blast design.

204.45 Vibration Control and Monitoring

204.45.01 Vibration Control - The Contractor shall use blasting methods designed to limit the intensity of ground vibrations originating within the excavation limits. When blasting near buildings, structures, wells, utilities or other works that may be subject to damage from blast induced ground vibrations, the ground vibrations shall be controlled using properly designed delay sequences and allowable charge weights per delay.

204.45.02 Charge Weight Per Delay - Allowable charge weights per delay shall be based on vibration levels that will not cause damage. The Quality Manager may monitor vibration levels at the blast site by requesting trial blasts to determine actual vibration levels reached during blasting.

204.45.03 Monitoring - Monitoring shall meet ISRM standards. Whenever vibration damage to adjacent structures is possible, the Contractor shall monitor each blast with approved seismograph(s) located between the blast area and the closest structure(s) subject to potential blast damage. The geophone shall be placed as close as possible to the structure(s) but not directly above the structure(s). The seismograph(s) shall be set to record particle velocity, accelerations, and frequency in the range generally found with controlled blasting. The peak particle velocity shall be calculated as the maximum vector sum of three mutually perpendicular components of vibration. All components and peak particle velocity shall be recorded.

204.45.03 Vibration Limits - Peak particle velocity, accelerations and frequency shall not be allowed to exceed

the safe limits of the nearest structure subject to potential vibration damage. The Contractor shall employ a qualified vibration specialist to establish the safe vibration limits.

204.45.04 Interpretation - The vibration specialist shall interpret the seismograph records to ensure that the seismograph data is effectively used in the control of the blasting operations with respect to the existing structures.

204.46 Air Concussion and Noise Control

204.46.01 General - When requested by the Quality Manager, an air concussion monitoring system shall be installed in a representative location between the blasting area and the nearest structure subject to potential blast damage or annoyance. The equipment used to make the air concussion measurements shall be the type specifically manufactured for that purpose.

204.46.02 Monitoring - The air concussion monitoring system shall be set to record air overpressure on the linear setting. Human annoyance (A weight setting) and human ear response (C weight setting) shall also be recorded.

204.46.03 Overpressure - Peak overpressure shall be controlled using appropriate blasthole patterns, detonation systems and stemming to prevent venting of blasts, and to minimize air concussion and noise levels produced by the blasting operations. The contractor shall use a qualified vibration specialist to establish safe overpressure limits.

204.47 Excavation Backslope Stabilization

204.47.01 Stabilization - The excavation backslope shall be stabilized by the Contactor as recommended by the Contractor's specialist Geotechnical Engineer and the Quality Manager, during or upon completion of the excavation of each lift. Drilling of the next lift shall not proceed until stabilization has been completed. Drilling of the next lift shall not proceed until all concerns about stability raised by the Contractor's geotechnical engineer, the Quality Manager and the Ministry Representative are addressed by the stabilization work, and Stabilization shall be completed before any base material is placed upon the subgrade.

204.47.02 Stabilization Methods - Stabilization methods include scaling, trimming, the application of rock bolts, shotcrete, slope mesh, drains or other stabilization techniques recommended by the Contractor's geotechnical engineer and authorized by the Quality Manager.

204.48 Special Use of Excavated Rock - Excavated rock of suitable quality required for walls, riprap, paving, or other special use shall be sorted and deposited in stockpiles if so requested by the Contract.

Stabilization shall be at the Contractor's expense.

PROPOSED BLAST DESIGN

File #: _____

PROJECT NO. _____

(TO BE SUBMITTED NOT LESS THAN 1 DAY BEFORE DRILLING)

GENERAL:

DATE PREPARED_____
CONTRACTOR'S NAME _____ PROPOSED BLAST DATE _____
BLASTER'S NAME _____ BLAST # _____
BLASTER'S CERTIFICATE NUMBER _____

HIGHWAY#_____
BLAST LOCATION AT: _____

UTILITY PROTECTION OR STANDBY REQUIRED: (CIRCLE) YES NO

SITE DETAILS:

HEIGHT ABOVE HWY _____ (m) SLOPE ANGLE _____ (°) SLOPE LENGTH _____ (m)
ROCK TYPE _____
DISTANCE TO NEAREST STRUCTURE (UTILITY)_____ (m)

ANTICIPATED DRILLING DETAILS:

PROPOSED NO. OF BACKLINE HOLES_____ PROPOSED APPROX. NO. OF HOLES _____
PROPOSED AVG. DEPTH _____ (m) HOLE DIA. _____ (mm)
PROPOSED NO. OF ROWS _____ APPROX. HOLE SPACING _____ (m) BURDEN _____ (m)
PROPOSED MAXIMUM DEPTH _____ (m) PROPOSED HOLE INCL.: (CIRCLE) VERTICAL HORIZ. VARIABLE
PROPOSED TOTAL DEPTH _____ (m)

PROPOSED BLAST DETAILS:

EXPLOSIVE TYPE _____ EXPLOSIVE SIZE _____mm by _____mm
APPROX. NUMBER OF CARTRIDGES _____ APPROX. TOTAL WEIGHT _____ (kg)
PROPOSED NUMBER OF DELAYS _____ DELAY TYPE AND LENGTH _____ (ms)

INITIATION DEVICE: (CIRCLE) E.B. CAPS SAFETY FUSE OTHER (SPECIFY) _____
BLASTING MACHINE: TYPE _____ CAPACITY _____

APPROX. VOLUME OF ROCK TO BE BLASTED _____ (m^3)

PROPOSED BLAST SKETCH:

SHOW THE FOLLOWING:

- SCALE
- PLAN VIEW
- CROSS-SECTION
- ROCK GEOMETRY
- APPROX. HOLE LOCATIONS
- APPROX. HOLE DEPTH
- PROPOSED TIE-IN PATTERN
- ROW BY ROW DELAYS
- DETONATION DIRECTION
- HIGHWAY LOCATION
- UTILITY LOCATION
- NORTH ARROW

PLAN VIEW:	CROSS-SECTION VIEW:

BLASTING CONSULTANT NAME _____ SIGNATURE_____ COMPANY_____

BLASTER'S SIGNATURE _____ COMPANY_____

RECEIVED BY QUALITY MANAGER (SIGNATURE) _____

(Use Additional Sheets if Necessary)

AS-BUILT BLASTING RECORD

File #: _____

PROJECT NO. _____

(TO BE SUBMITTED NOT MORE THAN 1 DAY AFTER EACH BLAST)

GENERAL:

DATE PREPARED_____
CONTRACTOR'S NAME _____ ACTUAL BLAST DATE _____
BLASTER'S NAME _____ BLAST # _____
BLASTER'S CERTIFICATE NUMBER _____

HIGHWAY#_____
BLAST LOCATION AT: _____

UTILITY PROTECTION OR STANDBY USED: (CIRCLE) YES NO TYPE OF UTILITY PROTECTION _____

SITE DETAILS:

HEIGHT ABOVE HWY _____ (m) SLOPE ANGLE _____ (°) SLOPE LENGTH _____ (m)
ROCK TYPE _____
DISTANCE TO NEAREST STRUCTURE (UTILITY)_____ (m)

DRILLING DETAILS:

NUMBER OF BACKLINE HOLES_____ TOTAL NUMBER OF HOLES _____
AVERAGE DEPTH _____ (m) HOLE DIAMETER _____ (mm)
NUMBER OF ROWS _____ HOLE SPACING _____ (m) BURDEN _____ (m)
MAXIMUM DEPTH _____ (m) HOLE INCLINATION: (CIRCLE) VERTICAL HORIZ. VARIABLE
TOTAL DEPTH _____ (m)

BLAST DETAILS:

EXPLOSIVE TYPE _____ EXPLOSIVE SIZE _____mm by _____mm
TOTAL NUMBER OF CARTRIDGES _____ TOTAL WEIGHT _____ (kg)
TOTAL NUMBER OF DELAYS _____ DELAY TYPE AND LENGTH _____ (ms)

INITIATION DEVICE: (CIRCLE) E.B. CAPS SAFETY FUSE OTHER (SPECIFY) _____
BLASTING MACHINE: TYPE _____ CAPACITY _____

VOLUME OF ROCK BLASTED _____ (m^3)

BLAST SKETCH:

SHOW THE FOLLOWING:	PLAN VIEW:	CROSS-SECTION VIEW:
SCALE PLAN VIEW CROSS-SECTION ROCK GEOMETRY HOLE LOCATIONS HOLE DEPTH TIE-IN PATTERN ROW BY ROW DELAYS DETONATION DIRECTION HIGHWAY LOCATION UTILITY LOCATION NORTH ARROW		

BLASTER'S SIGNATURE _____ COMPANY_____

RECEIVED BY QUALITY MANAGER (SIGNATURE) _____

(Use Additional Sheets if Necessary)

SECTION 205

RIPRAP

205.01 General - This Section covers the protection by riprap of embankments and channels at the locations and of the type and class shown on the Design or required by the Contract.

Work within any watercourse shall generally be carried out in conformity with the environmental protection provisions to the satisfaction of the Designer, the Quality Manager, and the Ministry Representative.

205.02 Material - Rock shall be hard durable angular quarry rock of a quality that will not disintegrate on exposure to water or the atmosphere. The gradation of rock sizes (mass in kg) in each class of riprap, as specified or directed, shall conform to Table 205-A.

Rocks generally shall be evenly graded, approximately the stipulated sizes, and individual rocks shall have a thickness greater than one-third their length and none shall have a mass greater than five times that of the specified class mass.

For visual comprehension only, Table 205-B indicates the approximate average dimension of an angular rock for each specified rock class mass.

205.03 Preparation - Areas to receive riprap shall be trimmed to a uniform surface and to the slope(s) indicated on the Design or as directed by the Designer.

Before rock placement commences, loose material shall be removed and minor pot-holes and hollows filled with selected materials well tamped-in to the approval of the Designer and the Quality Manager.

205.04 Foundations - To provide a stable foundation and protection against any undercutting, the riprap shall be thickened at the toe, laid horizontally to form an apron and/or keyed into the bed of the watercourse, all as indicated on the Design or as directed by the Designer.

205.05 Filter Blankets - Filter blanket material and placement, where required, shall be as specified in the Design or as directed by the Designer.

205.06 Loose Riprap - The controlled placement of rock of the class specified shall produce a rock mass of the nominal or required thickness over the area indicated. The rock shall be manipulated as necessary to provide mass stability and a regular surface with a minimum of voids.

205.07 Hand-Laid Riprap - Hand-laid riprap, normally Class 10 or 25, shall conform to the size, gradation and requirements set out in DBSS 205.02. Individual rocks too large to handle shall be manipulated for satisfactory setting and spacing.

At the toe of sloped riprap, a sufficient number of the larger rocks shall be placed to form a firm foundation

TABLE 205-A GRADATION OF ROCK SIZES IN EACH CLASS OF RIPRAP

CLASS OF RIPRAP (kg)	*NOMINAL THICK-NESS OF RIPRAP (mm)	ROCK GRADATION: PERCENTAGE LARGER THAN GIVEN ROCK MASS (kg)		
		85%	50%	15%
10	350	1	10	30
25	450	2.5	25	75
50	550	5	50	150
100	700	10	100	300
250	1000	25	250	750
500	1200	50	500	1500
1000	1500	100	1000	3000
2000	2000	200	2000	6000
4000	2500	400	4000	12000

* The minimum thickness of riprap, measured at right angles to the slope, for the class specified, shall be the nominal thickness stated.

TABLE 205-B APPROXIMATE AVERAGE DIMENSION OF AN ANGULAR ROCK FOR EACH SPECIFIED ROCK CLASS MASS

kg	10	25	50	100	250	500	1000	2000	4000
mm	200	300	350	450	600	800	1000	1200	1500

approximately 50% thicker than the required nominal riprap thickness. The remaining larger rocks shall be regularly spaced, at least one to every 2.5 m², when placing the general rock mass to the nominal or required thickness over the area indicated. Smaller rocks or spalls shall be well hammered in to fill the interstices and to form a closely massed regular surface.

Where riprap is required in two layers, the rocks shall be laid up and generally lap jointed between the regularly spaced larger rocks placed as through headers.

205.08 Grouted Riprap - Where grouted riprap is shown or required, the surfaces of the rocks shall be cleaned and wetted and the interstices filled with cement mortar, well rodded and pounded in for a minimum mortar depth of 300 mm or as otherwise detailed or required by the Designer. The mortar shall consist of one part Portland cement to three parts well-graded clean fine aggregate mixed to a proper consistency.

SECTION 206

ROCK BOLTS

DESCRIPTION

206.01 Scope - This Section covers the installation, by appropriate rock bolting methods, of anchored steel bars tensioned and locked off against face plates in the areas designated by the Designer. Unless otherwise stated below, all rock bolts shall be installed and tensioned to the rock bolt manufacturer's specifications.

206.02 Codes And Standards

CAN/CSA G164	Hot Dip Galvanizing of Irregularly Shaped Articles.
CAN/CSA G30.18	Billet Steel Bars for Concrete Reinforcement
CAN/CSA G40.21	Structural Quality Steels
CAN/CSA A23.2-9C	Compressive Strength of Cylindrical Concrete Specimens.

206.03 Quality Control - The Contractor shall provide quality control and quality assurance for rock bolt operations.

206.04 Submittals - The Contractor shall supply the following to the Ministry Representative:

206.04.01 Proposed Rock Bolt System - Provide documentation as follows for the proposed rock bolt system:

a) Rock Bolts - Type, dimensions and manufacturer of the bar, centralizers, face plate, washer (where applicable), and nut.

b) Resin - Published specifications showing name of material, manufacturer, graph of unconfined compressive strength development versus time, gel time, viscosity, shelf life, storage and handling requirements.

c) Grout - Published specifications showing name of material, manufacturer, 7 and 28 day unconfined compressive strength, composition, shelf life, storage and handling requirements.

d) Data Sheets - Upon request by the Ministry Representative, supply the manufacturer's product data sheets.

e) Properties of Steel - Upon request by the Ministry Representative, supply the physical and chemical properties for each lot or heat number of the steel.

f) Test Results - Upon request by the Ministry Representative, submit a certified statement from an independent testing laboratory of the physical dimensions and mechanical properties of the rock bolt bar proposed for use. This statement shall include:

- Minimum cross-sectional area of the threaded portion of the bar.
- Minimum cross-sectional area of the unthreaded portion of the bar.
- Yield strength of the bar.
- Ultimate tensile strength of the bar.
- Percent elongation of the bar at yield and ultimate strength.

g) Quick Setting Mortar - Name of material, manufacturer, 7 and 28 day unconfined compressive strength, and composition.

h) Sealing - Proposed method of sealing fractures and voids in drill holes to prevent loss of grout.

206.04.02 Calibration Certificates - Provide tensioning jack calibration certificates before commencing rock bolt installation. Calibration of jacks shall have been performed by an authorized testing agency not more than 30 days prior to rock bolt testing. The certificate shall show the relationship between gauge pressure and applied load. Pumps and jacks shall be paired for calibration.

206.04.03 Contractor's Rock Bolt Records - Provide daily rock bolt tensioning records within one day of each day's rock bolt operations. Records shall include contractor name, date of tensioning, weather, temperature, test jack identification number, pump identification number, name of person who tested the rock bolt, rock bolt location, rock bolt number, rock bolt length, resin or grout details, test start time, test end time, gauge reading for each minute of the creep test, and lock off load. For resin systems, records shall also include the start and end times of spinning the bar through the resin in the hole.

MATERIALS

206.11 Materials - Steel materials shall be hot-dip galvanized to CSA G164. All resin, grout and steel materials shall be the products of established manufacturers regularly engaged in the manufacture of rock bolt materials for at least five years. Materials shall meet the following additional requirements:

a) Rock Bolt Bars

- Steel hot-rolled Grade 517 MPa meeting CAN/CSA G30.18.
- Nominal bar diameter 22 mm unless otherwise

specified.
- Threadlike surface deformations for full length of bar and suitable for mechanical coupling.
- Cut-thread reinforcing bar not permitted.

b) Miscellaneous Hardware

- Steel hardware to be compatible in size and strength with rock bolt bars.
- Face plates to be square in plan and dome shaped in cross-section.
- Face plate size to be matched to bar size.
- Face plates to CAN/CSA G40.21 Grade 300W.
- Face plate dimensions 10 mm by 150 mm by 150 mm unless otherwise specified.
- Face plates date stamped after galvanizing on the side visible when installed with the current year (in the format YYYY) in numbers 10 mm high.
- Face plates slotted for grout tube if grout is used.

c) Resin

- Fast-set and slow-set resin in cartridge form.
- A minimum unconfined compressive strength when fully mixed and cured of 90 MPa, tested in accordance with CAN/CSA A23.2-9C.
- Encased in a plastic film that provides optimum resistance to moisture, and is easily ruptured to enable complete mixing during installation.
- Suitable thixotropic and viscous properties to permit adequate mixing of the resin components by rotation of the rock bolt bar and to contain the resin within the drill hole.
- Easily identifiable gel time and as recommended by the resin manufacturer.
- Reach 80% of its ultimate strength within a time interval equal to five times the gel time.
- Non-shrink after the gel time.
- Unaffected by mild acids or mild alkalis.
- Cartridge boxes labelled with the resin expiry date.

d) Grout

- Pre-mixed, unsanded, non-metallic, and non-shrink cementitious grout containing silica fume.
- Can be mixed to a flowable consistency, typically with a water to cement ratio of 0.35 or as specified by the grout manufacturer to achieve or exceed the required strength.
- Minimum 7 day compressive strength of 30 MPa and a minimum 28 day compressive strength of 40 MPa, tested in accordance with CAN/CSA A23.2-9C.
- Admixtures to be used according to the manufacturer's specifications and as authorized by the Ministry Representative.
- Calcium chloride accelerator is not permitted.

e) Mortar Pads

- Portland cement based.
- Quick setting.

CONSTRUCTION

206.31 Execution

206.31.01 General - The entire rock bolt system shall be stored under cover away from deleterious materials. All grease and other deleterious material shall be removed from the steel prior to rock bolt installation.

206.31.02 Site Preparation - Where rock bolts may be adversely impacted, rock removal above and around proposed rock bolt locations shall be completed before installation of rock bolts commences. Any minor rock scaling performed in conjunction with rock bolting shall be considered incidental to rock bolting.

206.31.03 Drill Holes

a) Location, Orientation and Depth - The location, direction, angle and depth of the holes will be dependent on field conditions. Prior to construction, the location, direction and depth shall be detailed by the Designer and the information provided to the Ministry Representative.

b) Hole Diameter - The diameter of the holes shall be suitable for the rock bolt system chosen. Where grout is used, the hole size shall be according to the rock bolt manufacturer's recommendations. Where resin is used, the hole size shall be according to the resin manufacturer's recommendations.

206.31.04 Cleaning - All water, grease, oil, cuttings and other deleterious materials shall be removed from finished holes by a water and/or air jet as required.

206.31.05 Sealing The Contractor shall take measures to seal fractures and voids in drill holes where necessary to prevent anchor grout loss.

206.31.06 Installation

a) General - Rock bolts shall be inserted (or rotated) into the drill holes and fully encapsulated in resin or grout to the drill hole collar. When resin is used, the bolt shall be advanced and rotated at a rate recommended by the resin manufacturer.

b) Anchorage Length - The anchorage length or bond zone shall be the last 1000 mm of the inserted end of the bar, unless otherwise specified.

c) Centralizers - If grout is used, centralizers on 3.0 m centres shall centralize the rock bolt in the drill hole before grout is placed. Centralizers shall be suitable for holes in rock and shall be specifically manufactured for rock bolt application.

d) Resin - Resin cartridges shall be installed as follows or as specified by the resin manufacturer:

- **Fast-Set Resin** - A sufficient number of fast-setting cartridges shall be placed in the bond zone at the bottom of the hole for the anchorage.

- **Slow -Set Resin -** A sufficient number of slow-setting cartridges shall be placed in the free stressing length between the bond zone and the collar of the hole.

e) Grout - Grout shall be prepared and placed as follows unless otherwise specified by the grout manufacturer:

- **Mixing -** Grout shall be mixed in a colloidal or high shear grout mixer according to the grout manufacturer's published instructions. Mixing paddles shall be slotted and perforated. Mixing time shall be not less than two minutes.

- **Batching -** All ingredients for the grout mix shall be batched by mass. Water shall be added to the drum first and dry ingredients afterwards. Grout shall not be re-tempered after initial mixing. Grout shall be placed immediately after mixing.

- **Grout Placement -** Grout shall be pumped using a grout tube extending to the bottom of the hole. The inserted end of the tube shall remain below the level of the grout in the hole to effect a continuous air free column as the grout level rises. Grout shall be placed quickly and continuously to avoid overworking, segregation, bleeding and disturbance of initial set. Grout that has stiffened due to delay in placing shall not be used in the work and shall be disposed of at an authorized location.

f) Resin and Grout - At the Contractor's option, both resin and grout may be used to surround a rock bolt as follows. Resin cartridges may be placed in the bond zone, and grout may be used in the free-stressing length.

g) Rock Face Preparation - The bearing surface shall be prepared to allow the face plate to be oriented within the limits recommended by the anchor manufacturer. If necessary, rock shall be chipped from around the face plate contact area.

h) Mortar Pad Construction - Mortar pads shall be constructed as required to ensure the bar is within 20° of a line perpendicular to the face plate. Mortar pads shall be installed where face plates deform excessively under load. The pad shall not crack or deform when loaded. Sufficient time shall be provided to allow pads to achieve sufficient bearing capacity prior to test-tensioning.

i) End Hardware Installation

- Nuts shall bear uniformly against the faceplate.
- The bolt extension beyond the nut shall be 100 mm ± 10 mm.

206.31.07 Tensioning - All rock bolts shall be test-tensioned and locked off following set-up (or curing) of the bond zone and before set-up of the free stressing length. Prior to testing, the bond zone grout and/or resin shall meet the strength specified by the anchor manufacturer. The following procedure applies to 22 mm diameter Grade 517 MPa steel bars. An alternative procedure may be specified for different bar sizes.

a) Equipment - Equipment required for tensioning shall be supplied by the Contractor and shall be of a size adequate to provide the required tension. A torque wrench shall not be used for tensioning.

b) Test-Tensioning and Creep Test - Rock bolts shall be test-tensioned to 186 kN (42,000 lb). The test load shall be held for 10 minutes for the creep test.

c) Lock-Off Tension - Rock bolts shall be locked-off to a design tension of 140 kN (31,500 lb) after testing.

d) Acceptance Criteria - During the creep test a load loss of greater than 10% of the load applied shall be indicative of anchorage failure. Creep movement at the anchor head shall not exceed 2 mm during the creep test. A replacement rock bolt shall be installed at the Contractor's expense where these criteria are not met.

206.31.08 Rock Bolt Evaluation - The Contractor shall implement a program of evaluation of rock bolts installed. The Contractor shall perform the following test procedure on randomly selected bolts. After locking off the anchor, the load shall be re-applied to determine the lift-off load. The lift-off load shall be the tension level at which the

anchor nut can be loosened by hand. Lift-off tests shall be performed on rock bolts chosen by the Designer to a minimum of 5% of the total number of rock bolts. One additional lift-off test on a different bolt shall be performed for each bolt whose lift-off load is not within 10% of the specified lock-off load. Following lift-off testing, all bolts shall be locked off as specified.

The Contractor shall also perform the above evaluation on additional bolts identified by the Ministry Representative, as part of the Ministry's audit program, on up to 2% of the rock bolts installed

File # _____

CONTRACTOR'S DAILY ROCK BOLT TESTING AND TENSIONING RECORD

PROJECT NO. _____

(TO BE SUBMITTED WITHIN 1 DAY AFTER EACH DAY'S ROCK BOLTING OPERATION)

Contractor Name _____

Bolt Type _____ Size _____

Resin/Grout Type _____ Anchorage Length _____ m

Test Jack Number(s) _____

Sample calculation for this grout:

Weight of each bag of grout (M) _____ kg

Volume of water added for each bag (V) _____ L

Actual Water / Cement Ratio (V/M) _____

Date _____

Weather _____

Temperature _____ °C

Name of Person who Installed and Tested the Rock Bolt	Zone	Area	Bolt Sequence Number for the Area	Bolt Length (m)	Bolt Rotation Time Through Resin		Test Start Time	Test Details: Record Gauge Reading of Each Minute										Lock Off Load (kN)	Test End Time	Grout Cube #	Certified Correct by Tester (Signature)
					Start	End		Elapsed Time (minute) 1	2	3	4	5	6	7	8	9	10				

Certified Correct: _____

(Contractor's Superintendent)

SECTION 207

SLOPE MESH FOR ROCK CUTS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

207.01 Scope - This Section applies to slope mesh structures which are installed to provide rockfall protection.

207.01.01 Rock slope scaling and tree removal shall be performed in conjunction with slope mesh installation. Rock slope scaling and tree removal shall be completed before the mesh support system is installed. The Designer shall detail trees to be removed.

207.01.02 Discarded mesh components, scaled rock, trees and debris generated by the slope meshing work shall be removed from the ditches and disposed of by the Contractor. Temporary storage of felled trees in ditches will not be permitted.

207.01.03 The contractor shall layout in the field all anchor and suspension cable locations for each section.
207.01.04 Field conditions may require final anchor, cable and mesh configurations to vary from the Design. All variations must be detailed by the Designer and provided to the Ministry Representative for information.

207.01.05 The area to be provided with slope mesh protection shall be divided into sections with a maximum width of 24 m. Each section shall have a separate suspension cable, end main anchors and end anchors. End main anchors of adjacent sections shall be positioned 200 mm apart. The gap between sections shall be closed with mesh.

207.01.06 The maximum mesh length shall be 80 m.

207.01.07 The maximum rock impact energies shall be as follows:

- 3 kJ above cut crest
- 30 kJ below cut crest

207.01.08 Overburden thickness may range from zero to over 1.5 m.

MATERIALS

207.11 Slope Meshing Materials - Unless otherwise specified, preparation and installation of materials shall be according to manufacturer's recommendations.

All components shall be hot-dip galvanized conforming to ASTM A 123 or ASTM A 153 or CSA G164 where appropriate. Damaged galvanizing shall be re-galvanized.

207.11.01 Mesh - Mesh shall be 11 gauge (2.95 mm dia.) hexagonal triple twist gabion type mesh. Mesh wire shall meet federal specification QQ-W-461g, possess soft tensile strength with a finish 5 class 3 zinc coating of not less than 260 g/m². The weight of zinc coating shall be determined by ASTM A 90. The coating shall withstand four one minute dips by the preece test, ASTM A 239. Mesh opening shall be hexagonal in shape and uniform in size measuring 80 mm by 100 mm.

207.11.02 Threadbar - Threadbar shall conform to CSA G30.18, grade 400 steel, manufactured by Dywidag Systems Int. (DSI) or authorized equivalent.

207.11.03 Eye nuts - Eye nuts shall be cast or manufactured eye nuts by DSI or authorized equivalent.

207.11.04 Cables - All cables shall be fibre core conforming to CSA G4. Cables shall be unspliced.

207.11.05 Thimbles - Thimbles shall meet FF-T-276B Type III (extra heavy G-414).

207.11.06 Clips - Clips shall meet FF-C-450 Type 1, Class 1 (G450 Crosby clip or authorized equivalent).

207.12 Grout

207.12.01 Grout for main anchors shall be Celtite Anchortite. Grout for auxiliary anchors shall be Celtite Lokset cartridges, cement grout or authorized equivalent.

207.12.02 Cement grout shall be Target 1118 or Ocean Microsil anchor grout or equivalent with W/C=0.35. Grout minimum 3 day and 28 day compressive strengths shall be 20 MPa and 40 MPa respectively, tested in accordance with CSA A23.2-1B.

207.13 Conformance Documents - Prior to installation the contractor shall supply documents of conformance to project specifications of all materials upon request.

CONSTRUCTION

207.31 Anchor Installation

207.31.01 Overburden shall be excavated to rock at anchor locations unless specified otherwise. The Contractor shall minimize disturbance of surrounding soil and rock when excavating. Cables shall not contact ground surface.

207.31.02 Anchor holes shall be a minimum of 1.5 times anchor diameter and in strong, competent rock. Anchors shall be centered in the hole and grouted. Anchors shall not be loaded within 3 days of grouting.

207.31.03 Main anchors shall be located at local high points where practicable to maximize clearance between suspension cable and ground surface. Main anchors shall be vertical and centered in the hole. Field conditions may require that the main anchor height above ground surface be reduced.

207.31.04 Auxiliary anchors shall be located to minimize potential for main anchor bending.

207.31.05 Main anchors may be substituted with a limited number of guy cables directly connected to the suspension cable where no suitable main anchor locations can be found. Guy cable anchors shall be located to maximize suspension cable elevation. See Section Y-Y, on Drawing SS SP207-02 and Detail B on Drawing SS SP207-03.

207.31.06 Auxiliary or guy anchor embedded length may require extension if weak rock conditions are encountered.

207.32 Soil Anchors - Soil anchors shall be used where required For soil anchor details see Drawings SS SP207-04 and SS SP207-05.

Concrete requirements for soil anchors:

- minimum compressive strength at 28 days = 30 MPa
- maximum nominal size of aggregate = 28 mm
- air content = 5± 1%
- slump = 55± 20 mm
- maximum w/c ratio by mass = 0.45

The contractor shall load test one overburden soil anchor to 10 kN to verify capacity.

207.33 Cable and Mesh Installation - Suspension, auxiliary, end, and guy cables shall be installed to nominal tension to remove slack before and after installing mesh.

A maximum of two horizontal mesh seams (200 mm overlap) shall be permitted along the entire mesh height. The upper mesh portion shall be between the slope and lower mesh at the overlap. The horizontal seam connections shall be similar to the vertical seams.

The mesh shall terminate 1500 mm above the highway edge of pavement elevation. The bottom of the mesh shall be evenly trimmed parallel with the highway elevation. The bottom of the mesh shall be bent to remove the curl.